⚡ PHYSICS CH.1 — LIGHT AND OPTICS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

📑 **Topics in this chapter:**

1. Spherical Mirrors: Image Formation & Position Rules

- Concave mirror (converging mirror) parallel rays ko focus par converge karta hai — isliye surya ki sharp image paper par burn kar sakti hai. *[Q5, Q39]*
- Convex mirror hamesha virtual, erect, diminished image banata hai, chahe object kahi bhi ho; image pole (0) aur focus (f) ke beech banega. *[Q39, Q79, Q125, Q221, Q259]*
- Concave mirror ki image formation positions (yaad rakho):
  - Object infinity par → image Focus (F) par, highly diminished, real, inverted. *[Q21, Q52, Q220, Q245, Q273]*
  - Object beyond C (2f) → image F aur C ke beech, diminished, real, inverted. *[Q36, Q47, Q69, Q251]*
  - Object at C (2f) par → image C (2f) par, same size, real, inverted. *[Q48, Q219]*
  - Object between C aur F → image beyond C, enlarged, real, inverted. *[Q66, Q216]*
  - Object at F → image at infinity, highly enlarged, real, inverted. *[Q52, Q220]*
  - Object between F aur Pole → image behind mirror, virtual, erect, enlarged. *[Q52, Q53, Q169, Q195]*
- Magnification analysis:
  - m negative (e.g., -0.5, -0.65, -1.38) → image real, inverted. Agar $|m| < 1$ → diminished; $|m| > 1$ → enlarged. *[Q12, Q89, Q107]*
  - m positive + $|m| > 1$ (e.g., +3.0) → image virtual, erect, enlarged (concave mirror ya convex lens dono de sakte hain). *[Q111]*
  - m positive + $|m| < 1$ (e.g., +1/2) → image virtual, erect, diminished. *[Q24]*

- Concave mirror normally converging hota hai, lekin jab object pole aur focus ke beech ho to reflected rays diverge karte hain (virtual image banane ke liye). *[Q130]*
- Convex mirror hamesha diverging hota hai, kabhi converge nahi kar sakta. *[Q127, Q130]*
- Ray Rules (Concave mirror):
  - Ray parallel to principal axis → reflects through focus. *[Q95, Q99, Q204]*
  - Ray through focus → reflects parallel to principal axis. *[Q95]*
  - Ray through C → wapas same path mein retraces (normal incidence, angle = 0°). *[Q2, Q13, Q15, Q40, Q59, Q103, Q138, Q150]*
  - Agar parallel beam principal axis ke parallel nahi hai, to wo principal focal plane par converge karegi. *[Q141]*
- Pole par oblique incidence wali ray apna path retrace nahi karti — ye sirf C se hokar jaane wali ray karti hai. *[Q124]*
- Mirror formula se rearrange karke: $f = (uv)/(u+v)$. *[Q112, Q132]*
- Graph: m vs v ka graph concave mirror ke liye straight line hota hai with slope $+1/f$, x-intercept $+f$, y-intercept -1. *[Q117]*
- Virtual + erect image sirf plane mirror aur concave mirror (jab object pole-focus ke beech ho) de sakte hain — convex mirror bhi hamesha virtual-erect deta hai; concave "hamesha" nahi (only special case). *[Q33]*
- Point source concave mirror ke Focus (F) par rakha ho to reflected rays parallel beam ban jaati hain (converging mirror ka reverse use). *[Q85]*
- Convex mirror ke focal length 1 m ho aur parallel beam incident ho, to reflected rays mirror ke peechhe 1 m distance wale point se diverge hote hue appear hoti hain (distance = focal length). *[Q96]*
- 'Converging mirror' naam concave mirror ko isliye milta hai kyunki ye parallel rays ko ek point (focus) par converge karta hai — inward curved (bulge center ki taraf) reflecting surface hoti hai. *[Q239, Q267]*

2. Spherical Mirrors: Formula, Sign Convention & Graphs

- Radius of Curvature: $R = 2f$; yaani P se C ki distance = 2 × focal length. *[Q4, Q23, Q92, Q121, Q225, Q244]*
- New Cartesian Sign Convention:
  - Object hamesha left side par rakha jaata hai. *[Q184, Q196]*
  - Right of origin (P) = positive; Left = negative. *[Q196]*
  - Above principal axis = positive; Below = negative. *[Q184, Q196]*
  - Concave mirror ka f = negative, R = negative. *[Q50, Q93, Q134, Q144, Q246]*
  - Convex mirror ka f = positive, R = positive. *[Q10, Q72, Q144, Q250]*
- Mirror Formula: $1/f = 1/v + 1/u$. *[Q173]*
- Magnification (m):
  - Formula: $m = -v/u = h_i/h_o$. *[Q200]*
  - Virtual/erect image ke liye m positive; real/inverted ke liye m negative. *[Q200, Q218]*
- ⚠️ Trap: Mirror ki focal length medium change hone par nahi badalti (jaise liquid mein dubone par), kyunki $f = R/2$ sirf geometric shape par depend karti hai. Lens ki focal length medium ke saath badalti hai. *[Q258]*
- Concave mirror banate waqt sphere ke slice ko is tarah polish karte hain ki reflecting (depression) surface centre of curvature ki taraf ho aur polish (silvering) opposite side par ho. *[Q83]*
- Radius of Curvature ki definition: Pole (P) aur Centre of Curvature (C) ke beech ki distance. *[Q110]*
- Agar spherical mirror aur thin lens dono ka focal length -10 cm diya ho, toh sign convention ke hisaab se dono concave honge (negative f sirf concave mirror/lens ka hota hai). *[Q194]*
- Spherical lens ke liye sign convention mein saari distances Optical Centre se measure ki jaati hain (mirror mein Pole se hoti hain). *[Q202]*

3. Lenses: Image Formation, Ray Rules & Power

- Convex lens (positive focal length) distant object ki image focus (f) par banata hai — lens se screen ki doori = f. *[Q8]*
- Convex lens ki image formation summary:
  - Object infinity → image at F2, highly diminished, real, inverted. *[Q7, Q30, Q255]*
  - Object beyond 2F → image F2 aur 2F2 ke beech, diminished, real, inverted. *[Q30, Q176]*
  - Object 2F par → image at 2F2, same size, real, inverted. *[Q7, Q190]*
  - Object F aur 2F ke beech → image beyond 2F2, enlarged, real, inverted. *[Q7, Q176]*
  - Object F par → image at infinity, highly enlarged, real, inverted. *[Q7, Q176]*
  - Object F aur Optical Centre ke beech → image same side par, virtual, erect, enlarged. *[Q7, Q197]*
- Concave lens hamesha virtual, erect, diminished image banata hai, chahe object kahi bhi ho infinity se O tak. *[Q9, Q70, Q175, Q198]*
  - Object infinity par → image at F, point-sized. *[Q175]*
  - Parallel ray concave lens se guzre to focus se diverge hote hue dikhti hai. *[Q180]*
  - Focus ki taraf jaane wali ray principal axis ke parallel niklegi. *[Q177]*
- Convex lens real inverted image twice the size → magnification = -2 (negative kyunki inverted). *[Q22]*
- Lens ke Optical Centre (O) se hokar guzarne wali ray bina kisi deviation ke seedhi jaati hai, aur yeh point hamesha principal axis par hota hai. *[Q56, Q142, Q189, Q238]*
- Principal axis wo imaginary line hai jo lens ke do foci (F1 aur F2) ko join karti hai. *[Q149, Q205]*
- Power (P) = 1/f (in meters); unit = Dioptre (D). *[Q139, Q266]*
  - Power positive (+2.0D) → convex lens. *[Q266]*
  - Power negative → concave lens. *[Q139]*
- Lens Formula: 1/f = 1/v - 1/u (ya 1/v - 1/u = 1/f). *[Q173]*
- Concave lens = thin in middle, thick at edges; Convex lens = thick in middle, thin at edges. *[Q161]*
- Lens banane ke liye material transparent hona chahiye; wood (lakdi) aur soil dono se lens nahi ban sakta kyunki dono opaque hain — glass, plastic, water se lens ban sakta hai. *[Q101, Q211]*
- Point source ko convex lens ya concave mirror ke focus par rakhne se dono hi parallel beam produce karte hain (dono converging optical devices hain). *[Q38, Q115]*
- Convex lens mein principal axis ke parallel incident ray refraction ke baad Principal Focus se hokar guzarti hai (dusri side). *[Q58, Q118]*

4. Plane Mirror: Properties & Lateral Inversion

- Plane mirror ki image hamesha: virtual, erect, same size, laterally inverted. *[Q224, Q228, Q237, Q256]*
- Lateral inversion = left side right mein aur right side left mein dikhta hai. *[Q228, Q260]*
- Object mirror se jitni doori par ho, image utni hi doori peechhe hota hai. *[Q224]*
- Agar mirror mein apna reflection chhota dikhe → convex mirror hai; same size → plane mirror; bada jab paas ho → concave mirror. *[Q259]*

5. Reflection Basics, Laws & Terminology

- Pole (P) = reflecting surface ka geometric centre; surface par hota hai. *[Q158, Q204, Q263]*
- Centre of Curvature (C) = us hollow sphere ka centre jiska mirror ek part hai. *[Q207, Q270]*
- Principal Focus (F) = wo point jahan principal axis ke parallel rays converge karte hain (concave) ya diverge hote

hue dikhte hain (convex). *[Q119, Q230, Q272]*

- Focal Length (f) = P se F ki distance; f = R/2. *[Q183, Q254]*
- Aperture = reflecting surface ka diameter; kitna light capture hoga ye decide karta hai. *[Q271]*
- Principal axis = wo imaginary straight line jo Pole, Focus aur Centre of Curvature teeno se hokar guzarti hai, dono taraf extended hoti hai — curved mirrors ke liye bhi ye hamesha straight line hoti hai (kabhi curved nahi hoti). *[Q143, Q145]*
- Spherical mirror par kisi bhi point se C ko join karne wali line normal hoti hai. Isliye ray C se hokar jaaye to angle of incidence = 0°, aur reflected ray bhi 0° par wapas jaati hai. *[Q2, Q13, Q15, Q40, Q59, Q103, Q138, Q150]*
- Principal axis par pole par ray incident ho to angle of incidence wahi hota hai jo principal axis se ban raha ho (kyunki principal axis normal hota hai pole par); oblique incidence wali ray angle i = angle r follow karte hue reflect hoti hai. *[Q28, Q32, Q82]*
- Laws of reflection (angle i = angle r, same plane mein) sabhi mirrors (plane, concave, convex) ke liye valid hain — jaise 30° incidence par reflection bhi 30° hi hota hai. *[Q35, Q247, Q269]*
- Spherical mirror ke surface par kisi bhi point ka C se distance hamesha baraabar hota hai (radius constant), isliye r1 = r2 = r3 = r4. *[Q19]*
- Pole aur Centre of Curvature dono hamesha spherical mirror ke principal axis par hote hain; point of incidence kahi bhi ho sakta hai. *[Q77]*
- Shiny/smooth surface se light regular reflection karta hai (specific direction mein), rough surface se diffused reflection hota hai (sabhi directions mein). *[Q222]*
- Light hamesha straight line (rectilinear path) mein travel karta hai — isi property ki wajah se opaque objects ke path mein aane par shadow banta hai. *[Q164, Q170]*

6. Refraction through Slab, Prism & Interfaces

- Refraction = ek medium se doosre medium mein jaane par light ka mudna. Glass slab mein: incident ray aur emergent ray parallel hoti hain (kyunki do bar opposite direction mein refraction hota hai). *[Q3, Q20, Q29, Q193]*
- Rarer (air) → Denser (glass ya water): light normal ki taraf muddta hai → angle of refraction < angle of incidence, speed slows down. Example: air se water (RI=1.33) mein jaate waqt 30° incidence par refraction angle 30° se kam banega. *[Q37, Q55, Q68, Q74, Q131, Q178, Q212, Q235, Q306]*
- Denser (glass) → Rarer (air): light normal se door muddta hai → angle of refraction (emergence) > angle of incidence inside glass, speed speeds up; light denser se rarer medium mein jaate waqt speed badhti hai isliye normal se door mudti hai. *[Q37, Q55, Q68, Q71, Q74, Q88, Q201, Q217, Q233, Q261]*
- Glass slab ke liye: angle of incidence = angle of emergence hota hai. *[Q29]*
- Prism mein angle of deviation = incident ray aur emergent ray ke beech ka angle. Formula: $\delta = (i + e) - A$. *[Q16, Q26, Q86]*
- Prism mein angle of incidence = angle of emergence sirf minimum deviation ki position par hota hai, hamesha nahi. *[Q62, Q122]*
- Prism mein light 2 baar refract hoti hai — pehli baar air se glass mein enter karte waqt, dusri baar glass se air mein exit karte waqt. *[Q105]*
- First law of refraction: incident ray, refracted ray, aur normal teeno same plane mein hote hain. *[Q80, Q247, Q306]*
- Normal se incidence par (0°) → refraction angle bhi 0°, dono rays ke beech angle 0°; direction change nahi hota sirf speed badalti hai. *[Q6, Q61, Q114]*
- Glass se hawa mein jaane par refracted ray 30° se zyada angle par bend hoti hai aur incident ray ke same plane mein rehti hai. *[Q68]*
- Agar light medium A se B mein ja kar normal se door mudd rahi hai, toh medium B rarer hai, isliye relative RI < 1 (less than unity). *[Q113]*
- RI badhne par deviation bhi badhta hai — kyunki light zyada sharply bend hoti hai. *[Q109]*
- Snell's Law: sin i / sin r = constant = RI (refractive index). *[Q151]*

- Medium ka light ko refract karne ki capacity uske optical density se measure hoti hai (mass density alag cheez hai). *[Q157]*
- Ordinary gases (jaise air) mein dispersion nahi dikhta kyunki unka RI ≈ 1, toh alag-alag wavelengths ki speed almost same hoti hai (≈ c). *[Q102]*
- Mirage (registan mein paani ka bhram) — total internal reflection ki wajah se hota hai (garam ground ke upar hot air rare medium ban jaata hai). *[Q156]*
- Pencil paani mein mudi hui (bent) dikhti hai / lemon bada dikhta hai — refraction ki wajah se. *[Q91, Q126, Q147, Q268]*
- Swimming pool ka floor kam gehra (shallower) dikhta hai — refraction se bottom ka virtual image upar shift hota hai. *[Q106]*

7. Speed of Light, Refractive Index & Medium Properties

- Speed of light in vacuum = 3×10^8 m/s — sabse zyada possible speed. *[Q232, Q262]*
- Vacuum mein speed max hai kyunki RI = 1 (exactly unity). *[Q232]*
- Absolute refractive index kisi bhi medium ka hamesha 1 se zyada hota hai (formula: $n = c/v$, aur $v < c$). *[Q31, Q234]*
- RI formula: $n = c/v$ = speed of light in vacuum/air ÷ speed in medium (also written $\mu = c/v$). *[Q100, Q136, Q146, Q186, Q203]*
- RI values yaad rakho:
  - Air ≈ 1.0003 (sabse kam). *[Q187, Q248, Q249]*
  - Ice = 1.31, Water = 1.33, Kerosene = 1.44, Benzene = 1.50, Rock Salt = 1.54, Glass ≈ 1.5, Diamond = 2.42. *[Q210, Q227, Q187]*
  - Order: Ice < Kerosene < Benzene < Rock Salt. *[Q210]*
- Refractive index aur speed ka relation: $n \propto 1/v$. Jitna zyada RI, utni kam speed. Example: RI 1.47 ka matlab uss medium mein speed vacuum ki speed se 1.47 guna kam hai. *[Q45, Q63, Q100, Q186]*
  - Example: RI 1.31, 1.46, 1.65 → speed order: V1 > V2 > V3. *[Q45, Q63]*
- Light paani se hawa mein jaane par uski speed badal jaati hai, lekin frequency aur colour same rehte hain. *[Q128]*

8. Dispersion of Light (VIBGYOR)

- White light prism se guzre to 7 colours (VIBGYOR) mein split hoti hai — isse dispersion kehte hain. Discovery: Isaac Newton (1666). Spectrum ke do extreme ends Violet aur Red hote hain (Violet shortest wavelength, Red longest wavelength). *[Q17, Q78, Q137, Q153, Q163, Q185, Q209]*
- Bending inversely proportional to wavelength hai:
  - Violet (shortest $\lambda \approx 400$ nm) → sabse zyada muddta hai, sabse zyada RI, sabse kam speed in glass. *[Q17, Q27, Q42, Q57, Q65, Q94, Q133, Q208]*
  - Red (longest $\lambda \approx 700$ nm) → sabse kam muddta hai, sabse kam RI. *[Q17, Q34, Q49, Q90, Q199, Q252]*
- RI order example: µViolet > µGreen > µYellow ya µBlue > µGreen > µYellow ya full order µV > µI > µB > µG > µY > µO > µR. *[Q42, Q57, Q73, Q97, Q98]*
- Different colours ka glass mein speed alag-alag hota hai — isliye alag angles par refract hote hain. *[Q34, Q64, Q185]*
- Dispersion surface AB par shuru ho jaata hai, kyunki har colour ka RI alag hota hai; surface AC par bhi alag-alag refraction angles hote hain. *[Q64, Q108]*
- Prism ko upright rakha jaye to spectrum mein Red top par (least bent) aur Violet bottom par (most bent) hota hai. *[Q133]*
- Rainbow ka formation: RRC key ke hisaab se refraction + scattering + dispersion ka combination (though traditionally total internal reflection bhi padhaya jaata hai). *[Q120, Q171]*

9. Scattering of Light & Sky Colour

- Scattering = light ka particles se takrakar alag direction mein bikharna. *[Q44, Q104, Q123]*
- Particle SIZE pe depend karta hai. *[Q182]*
- Fine particles (dust, gas molecules) → shorter wavelengths (blue) ko zyada scatter karte hain. Isliye sky blue dikhta hai. *[Q44, Q51, Q104, Q123, Q162]*
- Bade particles (clouds, milk, water droplets) → sabhi colours ko almost equally scatter karte hain (Mie scattering), isliye white dikhte hain. *[Q43, Q67, Q104, Q214]*
- Red light ki wavelength sabse zyada hoti hai, isliye fog/smoke se sabse kam scattered hota hai — isliye danger signals mein red use hota hai. *[Q49, Q181]*
- Red sunset/sunrise ka reason bhi scattering hai — atmosphere se hokar guzarne mein blue light scatter ho jaati hai, bachi hui red light aankhon tak pahunchti hai. *[Q84, Q160]*
- ⚠️ Trap: Atmosphere na hone par (space mein/outer space) sky black dikhega kyunki scattering karne wale particles hi maujood nahi hain, isliye scattering hoti hi nahi. *[Q51, Q81, Q243]*

10. Atmospheric Refraction & Twinkling

- Atmospheric refraction: atmosphere ki different layers (temperature/density) ki wajah se light ka bend hona. *[Q41, Q46, Q54, Q60, Q75, Q129]*
- Stars twinkle karte hain kyunki unki apparent position actual position se thodi upar (higher) hoti hai aur continuously change hoti rehti hai. *[Q11, Q46, Q60, Q129]*
- Planets twinkle nahi karte kyunki ye extended source hain (point source nahi), intensity variation noticeable nahi hota. *[Q14]*
- Sunrise/sunset mein sun ka disk chapta (flattened) dikhta hai aur hum sun ko 2 minutes pehle dekh paate hain (actual sunrise se) aur 2 minutes baad tak (actual sunset ke baad) — atmospheric refraction ki wajah se. *[Q41, Q54, Q168]*
- Aag ke upar garam hawa mein objects waver/shimmer karte hain — yeh bhi atmospheric refraction hai. *[Q75]*

11. Human Eye, Vision Defects & Corrective Lenses

- Eye ka lens (crystalline lens) + cornea milkar light ko refract karke retina par focus karte hain. *[Q188]*
- Retina par image real aur inverted banta hai; brain use process karke upright samajhta hai. *[Q242]*
- Iris pupil ka size control karke light ki quantity regulate karti hai. *[Q223]*
- Ciliary body lens ki shape change karke focal length adjust karti hai (accommodation). *[Q229]*
- Accommodation = eye lens ki ability to adjust focal length so that near aur distant objects dono retina par focus ho sakein. *[Q155]*
- Least distance of distinct vision = 25 cm normal healthy eyes ke liye. *[Q155]*
- Myopia (Nearsightedness) = distant objects blurry; image retina ke aage banta hai. Cause: eyeball ka elongation ya eye lens ki excessive curvature. Correction: concave lens. *[Q152, Q236]*
- Hypermetropia (Farsightedness) = near objects blurry; image retina ke peechhe banta hai. Correction: convex lens. *[Q152, Q241]*
- Presbyopia = dono problems ek saath; correction: bifocal lens (myopia + hypermetropia dono ke liye). *[Q213]*
- Human eye ki image formation refraction principle par based hai. *[Q160]*
- Chashma / spectacles refraction principle par kaam karte hain. *[Q116]*

12. Real-Life Applications of Mirrors & Lenses

- Concave mirror uses:
  - Torches, searchlights, vehicle headlights (bulb at focus → parallel beam). *[Q166, Q257, Q264]*
  - Shaving mirrors (virtual erect magnified image jab paas ho). *[Q172]*
  - Dentist mirrors (virtual erect enlarged teeth image). *[Q140, Q154, Q179, Q215]*
  - Solar furnaces (sunlight ko focus par converge). *[Q140, Q191]*
  - Dish antennas (receiver at focus). *[Q87]*
- Convex mirror uses:
  - Rear-view mirrors (wider field of view, virtual erect diminished). *[Q140, Q167]*
  - Security mirrors in stores/parking lots. *[Q240]*
  - Street lights mein as reflector taaki light zyada area mein spread ho. *[Q140]*
- Plane mirror uses:
  - Looking glasses, periscopes, kaleidoscopes. *[Q167]*
- Magnifying glass = convex lens of short focal length. *[Q231]*
- Magic mirror concept: Head bada dikhne ke liye concave (enlarged), body same size ke liye plane, legs chhote dikhne ke liye convex mirror. *[Q25, Q148]*
  - Reverse order (head same, body chhota, legs bada) → Plane, Convex, Concave. *[Q148]*
- Photolysis (light se hone wala decomposition reaction, jaise photosynthesis mein water ka breakdown) light-driven chemical process ka example hai — GK crossover fact jo light ke practical/chemical effect ko test karta hai. *[Q206]*

13. ⚠ Traps / Key Differences Summary (MCQ Q1–Q273)

- ⚠ Trap: Concave mirror converging hota hai lekin iski focal length negative hoti hai; convex mirror diverging hota hai lekin focal length positive. Ulta mat samajhna! *[Q5, Q10, Q18, Q50, Q72, Q76, Q144, Q246, Q250]*
- ⚠ Trap: Convex mirror hamesha virtual, erect, diminished image banata hai — enlarged kabhi nahi. *[Q39, Q125, Q135, Q140, Q221, Q259]*
- ⚠ Trap: Concave mirror mostly real inverted image banata hai, lekin jab object pole aur focus ke beech ho to virtual, erect, enlarged banta hai. *[Q130, Q135, Q169, Q195]*
- ⚠ Trap: Mirror ki focal length medium change se nahi badalti; lens ki badalti hai. *[Q258]*
- ⚠ Trap: Prism mein angle of incidence = angle of emergence sirf minimum deviation par hota hai, hamesha nahi. *[Q62, Q122]*
- ⚠ Trap: Glass slab mein incident aur emergent ray parallel hoti hain, prism mein nahi. *[Q29, Q193]*
- ⚠ Trap: Object concave mirror ke focus par ho to image infinity par real inverted banegi (highly enlarged), virtual nahi. *[Q52, Q220]*
- ⚠ Trap: Fine particles → blue scatter; Large particles → red scatter; Very large → white (equal). *[Q43, Q44, Q67, Q104, Q214]*
- ⚠ Trap: Setting sun ka red colour = scattering hai, refraction nahi. *[Q160]*
- ⚠ Trap: Rainbow formation: RRC key ke mutabiq refraction + scattering + dispersion (traditional TIR alag concept hai). *[Q120, Q171]*
- ⚠ Trap: Normal incidence (0°) par direction nahi badalti, sirf speed badalti hai. *[Q114]*
- ⚠ Trap: Optical centre se guzarne wali ray un-deviated rehti hai; principal focus se hokar nahi. *[Q142, Q189, Q238]*
- ⚠ Trap: Magnification +3.0 (positive) ka matlab virtual erect enlarged — convex lens ya concave mirror dono de sakte hain. *[Q111]*
- ⚠ Trap: Sign convention mein right of origin = positive, left = negative. *[Q196]*
- ⚠ Trap: Angle with surface vs angle with normal confusion — scientific answer alag ho sakta hai official key se

(jaise Q192 mein 30° surface angle par official answer 30° tha, jabki scientifically 60° hona chahiye tha). *[Q192]*

- ⚠ Trap: Real image = actual intersection of rays; virtual = apparent intersection. *[Q253]*
- ⚠ Trap: Moon non-luminous hai (sunlight reflect karti hai); Sun, bulb, candle, firefly luminous hain. *[Q265, Q226]*
- ⚠ Trap: TV remote mein Infrared (IR) light use hoti hai, visible nahi. *[Q174]*
- ⚠ Trap: LASER = Light Amplification by Stimulated Emission of Radiation. *[Q165]*
- ⚠ Trap: Solar power systems Photovoltaic effect par kaam karti hain. *[Q159]*

14. Numerical: Mirror Formula (Image/Object Distance)

- Mirror Formula: $1/f = 1/u + 1/v$ (yaad rakho: $uv/(u+v) = f$). *[Q274, Q275, Q282, Q293, Q299, Q302, Q304, Q323, Q335, Q339, Q340, Q351, Q361]*
- Convex mirror ke liye f hamesha positive; image behind mirror banega isliye v positive. Example: f=+2m, u=-3m → $1/v = 1/2 + 1/3 = 5/6$ → v = +1.2 m (behind mirror). *[Q275]*
- Concave mirror ke liye f aur v (real image ke liye) negative hote hain. Example: f=-20cm, v=-60cm → $1/u = -1/20 + 1/60 = -2/60$ → u = -30 cm. *[Q282]*
- Agar object focus aur C ke beech ho (between f aur 2f) → real inverted enlarged image beyond C par banega. *[Q286, Q324]*
- Agar object C se pare ho (beyond 2f) → real inverted diminished image F aur C ke beech. *[Q302, Q335]*
- Agar object infinity par ho → image focus par, highly diminished. Satellite wale question mein $u = -10^5$ m, hi = -5µm → f ≈ -0.5 m. *[Q346]*
- Newton's Formula for mirrors: Agar object focus se x door ho aur image focus se y door ho, toh $f^2 = xy$ ya $f = \sqrt{(xy)}$. Object distance = f + x, image distance = f + y. *[Q320]*
- Concave mirror mein parallel beam converge hoti hai focal point par, jahan f = R/2. Example: R=0.8m → f=0.4m, isliye rays mirror ke saamne 0.4 m par converge hogi. *[Q318]*
- ⚠ Trap: Shaving mirror question (Q326) mein official answer ne m = -1.33 (real image) maana, jabki logically virtual hona chahiye tha. Exam mein official key follow karo. *[Q326]*

15. Numerical: Lens Formula (Image/Object Distance)

- Lens Formula: $1/f = 1/v - 1/u$ (ya $1/v - 1/u = 1/f$). *[Q274, Q280, Q283, Q288, Q291, Q297, Q301, Q315, Q322, Q332, Q343, Q345, Q352, Q359]*
- Convex lens mein object 2F (2f) par ho → image bhi 2F par, same size, real inverted (m = -1). Example: u=-40cm, f=+20cm → v = +40 cm. *[Q301, Q322]*
- Convex lens mein object F aur 2F ke beech (jaise u=-12cm, f=8cm) → image beyond 2F par, enlarged. v = +24 cm, height = 2 cm (magnification -2). *[Q291]*
- Convex lens mein object F ke andar (u < f) → virtual erect enlarged image same side par. Example: P=+5D (f=20cm), u=-10cm → v = -20 cm (virtual). *[Q315]*
- Concave lens hamesha virtual, erect, diminished image banata hai. Example: f=-20cm, u=-10cm → v = -20/3 cm, m = +2/3 (<1). *[Q288]*
- Power = -10D (concave lens), u=-15cm → f=-10cm. Image hamesha virtual aur erect hi banegi. *[Q297]*
- Concave lens: f=-15cm, u=-30cm → $1/v = -1/15 + 1/30 = -1/30$ → v = -10 cm. *[Q352]*

16. Numerical: Power of Lens & Combined Power (Doublet)

- Power formula: P = 1/f (f in meters); unit = Dioptre (D). *[Q277, Q281, Q300, Q309, Q343, Q349, Q350, Q358, Q360]*

- Convex lens → P positive; Concave lens → P negative. *[Q277, Q309, Q358]*
- Example: f=25cm=0.25m → P = 1/0.25 = +4 D. *[Q350]*
- Example: f=50cm=0.5m → P = 1/0.5 = +2 D. *[Q360]*
- Example: P = -6.5 D → f = 1/(-6.5) = -0.1538 m = -15.38 cm. *[Q349]*
- Combined Power (lenses in contact): P = P1 + P2.
  - Convex (+5D) + Concave (-3D) → P = +2 D → f = 1/2 m = 50 cm. *[Q281]*
  - Convex (+2D) + Concave (-5D) → P = -3 D. *[Q300]*
  - Doublet: f1=+20cm (+5D), f2=-50cm (-2D) → P = +3 D, F = 33.3 cm. *[Q343]*
- Combined focal length formula: 1/F = 1/f1 + 1/f2 (with proper signs). *[Q300]*

17. Numerical: Magnification Problems (Height/Diameter)

- Mirror magnification: m = -v/u = hi/ho. *[Q274, Q286, Q289, Q302, Q304, Q307, Q316, Q325, Q326, Q327, Q328, Q335, Q337, Q339, Q340, Q351, Q354, Q362]*
- Lens magnification: m = v/u = hi/ho. *[Q274, Q280, Q283, Q288, Q291, Q314, Q315, Q322, Q332, Q345]*
- ⚠ Difference: Mirror mein m = -v/u (negative sign extra); Lens mein m = v/u (bina negative sign ke directly). *[Q313]*
- m negative → real inverted; m positive → virtual erect. *[Q200, Q274]*
- Example (Lens): u=-25cm, v=+30cm → m = 30/(-25) = -6/5 (real, inverted, enlarged). *[Q274]*
- Example (Mirror): u=-30cm, m=+1.5 → 1.5 = -v/(-30) → v = +45 cm (behind mirror, virtual). *[Q325]*
- Example (Concave mirror): u=-5cm, real image 3× → m=-3 → -3 = -v/(-5) → v = -15 cm. *[Q351]*
- Object height × |m| = Image height. Example: ho=6cm, m=2/3 → hi = 4 cm. *[Q332]*
- Diameter same formula: di = |m| × do. Example: do=10cm, m=-1 → di = 10 cm. *[Q322]*
- Convex mirror mein object 3.5f par ho → m = -2/5, do=10cm → di = 4 cm. *[Q335]*
- Christmas ball (convex mirror): D=10cm → R=5cm → f=+2.5cm. m=+0.5 → u = -2.5 cm. *[Q316]*

18. Numerical: Sign Convention, Focal Length & R = 2f

- R = 2f hamesha valid hai spherical mirrors ke liye. *[Q285, Q303, Q305, Q311, Q353, Q356]*
- Concave mirror: f = negative, R = negative. Example: R=12cm → f = -6 cm. *[Q295, Q296]*
- Convex mirror: f = positive, R = positive. Example: R=15cm → f = +7.5 cm; C = +30 cm. *[Q290, Q292, Q298, Q305]*
- Agar R double ho jaaye → f bhi double ho jaata hai (directly proportional). *[Q311]*
- PC (Pole to Centre) = 2 × PF (Pole to Focus). *[Q303]*
- FC (Focus to Centre) = f (kyunki PC=2f, PF=f, toh FC = 2f-f = f). *[Q305]*
- Formula derivation: 1/f = (u+v)/uv → f = uv/(u+v) → R = 2uv/(u+v). *[Q285]*
- f = -24cm (concave) → R = -48 cm. *[Q296]*
- f = +15cm (convex) → C = +30 cm. *[Q298]*

19. Numerical: Refractive Index, Snell's Law & Relative RI

- Snell's Law: n = sin i / sin r = c/v = v1/v2 (relative). *[Q276, Q278, Q284, Q287, Q294, Q306, Q308, Q310, Q319, Q329, Q330, Q334, Q341, Q344]*
- Relative RI: n_A w.r.t B = n_A / n_B = v_B / v_A. *[Q278, Q287, Q294, Q310, Q319]*

- Example: n_glass=1.52, n_turpentine=1.47 → n_glass w.r.t turpentine = 1.52/1.47 = 1.03. *[Q287]*
- Example: n_ice=1.31, n_glass=1.52 → n_glass w.r.t ice = 1.52/1.31 = 1.16. *[Q294]*
- Example: n_A=3/2, n_B=4/3 → n_A w.r.t B = (3/2)/(4/3) = 9/8. *[Q310]*
- Speed relation: $n \propto 1/v$. Jitna zyada RI, utni kam speed. *[Q278, Q319]*
  - n1=1.50, n2=1.36, n3=1.54 → Speed order: v2 > v1 > v3. *[Q319]*
- Air se water (n=1.3) mein: sin y / sin x = 1/1.3. Agar 30° angle se light air se paani mein enter kare (n=1.33), toh refracted angle 30° se kam banega aur ray incident ray ke same plane mein rahegi. *[Q284, Q306, Q334]*
- Optical fibre: n=1.45, i=22° → sin r = sin(22°)/1.45 ≈ 0.2583 → r ≈ 14.99°. *[Q329]*
- i=60°, n=√3 → sin r = sin(60°)/√3 = (√3/2)/√3 = 1/2 → r = 30°. *[Q308]*
- i=60°, n=3/2 → sin r = sin(60°)/(3/2) = (√3/2)×(2/3) = 1/√2 → r = 45°. *[Q344]*
- i=75°, r=40° → n_A w.r.t B = sin(40°)/sin(75°) ≈ 0.642/0.965 ≈ 0.67. *[Q341]*
- Wavelength method: μ = λ_air / λ_medium = 600/380 = 1.58. *[Q330]*
- ⚠ Trap: Agar ray normal se door mudd rahi hai (denser→rarer) → relative RI < 1 (0 < n < 1). *[Q312]*

20. Numerical: Speed of Light in Medium

- Formula: $v = c/n = (3 \times 10^8) / n$. *[Q278, Q279, Q333, Q347, Q355]*
- n=1.5 → $v = 3\times10^8/1.5 = 2 \times 10^8$ m/s. *[Q279, Q355]*
- n=1.33 (water) → $v = 3\times10^8/1.33 = 2.26 \times 10^8$ m/s. *[Q333]*
- n=2.42 (diamond) → $v = 3\times10^8/2.42 = 1.24 \times 10^8$ m/s. *[Q347]*
- Two media speeds given: $v_A=2\times10^8$, $v_B=2.25\times10^8$ → n_A w.r.t B = v_B/v_A = 2.25/2 = 1.125. *[Q278]*

21. Numerical: Apparent Depth & Real Depth

- Formula: n = Real Depth / Apparent Depth. *[Q342]*
- Real depth = 30 cm, n=1.33 → Apparent depth = 30/1.33 = 22.6 cm. *[Q342]*
- Object apparent depth par kam gehra (shallower) dikhta hai — yeh refraction ka result hai (denser medium ka bottom hamesha upar shift dikhta hai). *[Q106]*

22. Numerical: Critical Angle

- Formula: sin(Ic) = 1/n (denser→rarer, air ke liye n=1). *[Q336]*
- n=1.56 (flint glass) → sin(Ic) = 1/1.56 ≈ 0.64. *[Q336]*

23. Numerical: Graph-Based & Formula Derivation Problems

- 1/v vs 1/u graph (mirror): Equation 1/v + 1/u = 1/f.
  - y-intercept = 1/f. Agar y-intercept = 0.11 → f = 1/0.11 = 9.09 cm. *[Q338]*
- u vs 1/m graph (mirror): Straight line with slope +f, x-intercept -1, y-intercept +f. *[Q348]*
- m vs v graph: Straight line with slope +1/f, x-intercept +f, y-intercept -1 (concave mirror ke liye). *[Q117]*
- Dispersion: $D \propto 1/\lambda^3$. Agar λ' = 3λ → $D' = D/(3)^3 = D/27$. *[Q317]*

- Shift problems (screen & object): m=2 → v=3f; m=5 → v=6f; screen shift d=3f. Agar d given hai → f = d/3.
  - Q340: d=39cm → 3f=39 → f=13 cm. *[Q340]*
  - Q327: d=24cm → 3f=24 → f=8cm; object shift = f/6 = 4/3 cm. *[Q327]*
  - Q321: d → object shift = d/10. *[Q321]*

24. Numerical: Traps / Formula Cheat Sheet

- ⚠ Trap: Mirror magnification = -v/u; Lens magnification = v/u (bina negative sign). Difference mat bhoolna! *[Q313]*
- ⚠ Trap: Sign convention — Concave mirror/lens ka f negative; Convex mirror/lens ka f positive. *[Q282, Q292, Q295, Q296, Q298]*
- ⚠ Trap: Object hamesha left side par rakha jaata hai → u hamesha negative (New Cartesian). *[Q274, Q282]*
- ⚠ Trap: Angle with mirror surface vs angle with normal. Mirror se 20° ka matlab normal se 70° (i=70°, r=70°, total angle = 140°). *[Q357]*
- ⚠ Trap: Agar numerical mein object focus ke andar ho concave mirror mein → virtual erect enlarged image behind mirror (v positive). *[Q325]*
- ⚠ Trap: Mirror ki focal length medium change se nahi badalti; lens ki badalti hai. *[Q258]*
- ⚠ Trap: Official answer key kabhi-kabhi physics se alag ho sakti hai (jaise Q192, Q326, Q331, Q312). Solve karke official answer match karo. *[Q312, Q326, Q331]*
- ⚠ Trap: Convex mirror hamesha positive f aur positive v deta hai (virtual image behind mirror). *[Q275, Q289, Q304, Q323]*
- ⚠ Trap: Agar question mein diameter ya height pucha ho aur object ka size given ho → pehle m nikalo, fir multiply kar do. *[Q315, Q322, Q332, Q337, Q345, Q362]*